SMARTLENDER PROJECT REPORT

AI-Powered Credit Card Approval System

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PHASE 1: BRAINSTORMING & IDEATION

Project Title

SmartLenderAI-Powered Credit Card Approval System

Problem Statement

Banks and financial institutions receive thousands of credit card applications every day. A large share of these are rejected because of factors such as high existing loan balances, insufficient income, or excessive credit inquiries. Reviewing every application by hand is slow and prone to human error, which makes it difficult for a bank to scale its approval process while keeping decisions consistent across applicants.

Proposed Solution

SmartLender automates the credit card approval decision using machine learning. A predictive model is trained on historical applicant data covering income, employment, and credit history, and evaluates new applications in a manner comparable to a human credit analyst. Classification algorithms are compared - Logistic Regression, Random Forest, XGBoost, and Decision Tree - and the strongest performer is deployed through a Flask web application, with an optional IBM Watson Machine Learning pipeline available for cloud hosting.

Target Audience

- **Primary:** Bank credit analysts and compliance officers who need reliable decision support.
- **Secondary:** Prospective customers who want to check their eligibility before formally applying.

Technology Stack

- Python, NumPy, Pandas – Data handling
- Scikit-Learn, XGBoost – Model training
- Matplotlib, Seaborn – Exploratory analysis

- Flask – Web application
- IBM Watson Machine Learning – Optional cloud deployment

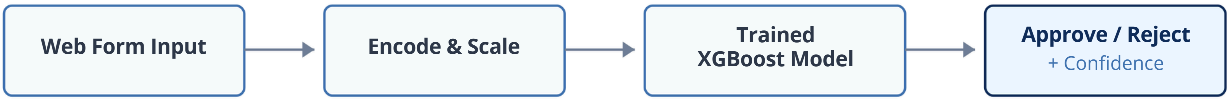

Figure 1: High-level prediction flow

PHASE 8: PROJECT DEMONSTRATION

Demo Scenario 1 – Automated Screening

A credit analyst enters an applicant's financial profile into the web app. The model returns an instant approve/reject prediction, letting the analyst prioritize which applications need closer review.

Demo Scenario 2 – High-Risk Identification

A compliance officer screens applicants with past-due loan records. The feature engineering pipeline converts multi-class payment status into a binary risk label, ensuring high-risk applicants are clearly flagged.

Demo Scenario 3 – Self-Service Eligibility Check

A prospective customer enters their own details before formally applying and receives an instant likelihood of approval, reducing the number of applications that would otherwise be rejected outright.

Terminal Execution

The screenshot below shows the Flask development server starting successfully and serving requests on local port 5000.

```
smartlender@dev-machine: ~/SmartLender/Flask

smartlender@dev-machine:~/SmartLender$ source .venv/bin/activate
(.venv) smartlender@dev-machine:~/SmartLender$ cd Flask
(.venv) smartlender@dev-machine:~/SmartLender/Flask$ python3 app.py
 * Serving Flask app 'app'
 * Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment.
Use a production WSGI server instead.
 * Running on http://127.0.0.1:5000
Press CTRL+C to quit
 * Restarting with watchdog (inotify)
 * Debugger is active!
 * Debuger PIN: 529-836-692
127.0.0.1 - - [02/Jul/2026 06:42:28] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [02/Jul/2026 06:42:32] "POST /predict HTTP/1.1" 200 -
```

Figure 5: Flask server running in terminal

Web Interface – Input Form

The applicant enters demographic and financial details through the web form shown below.

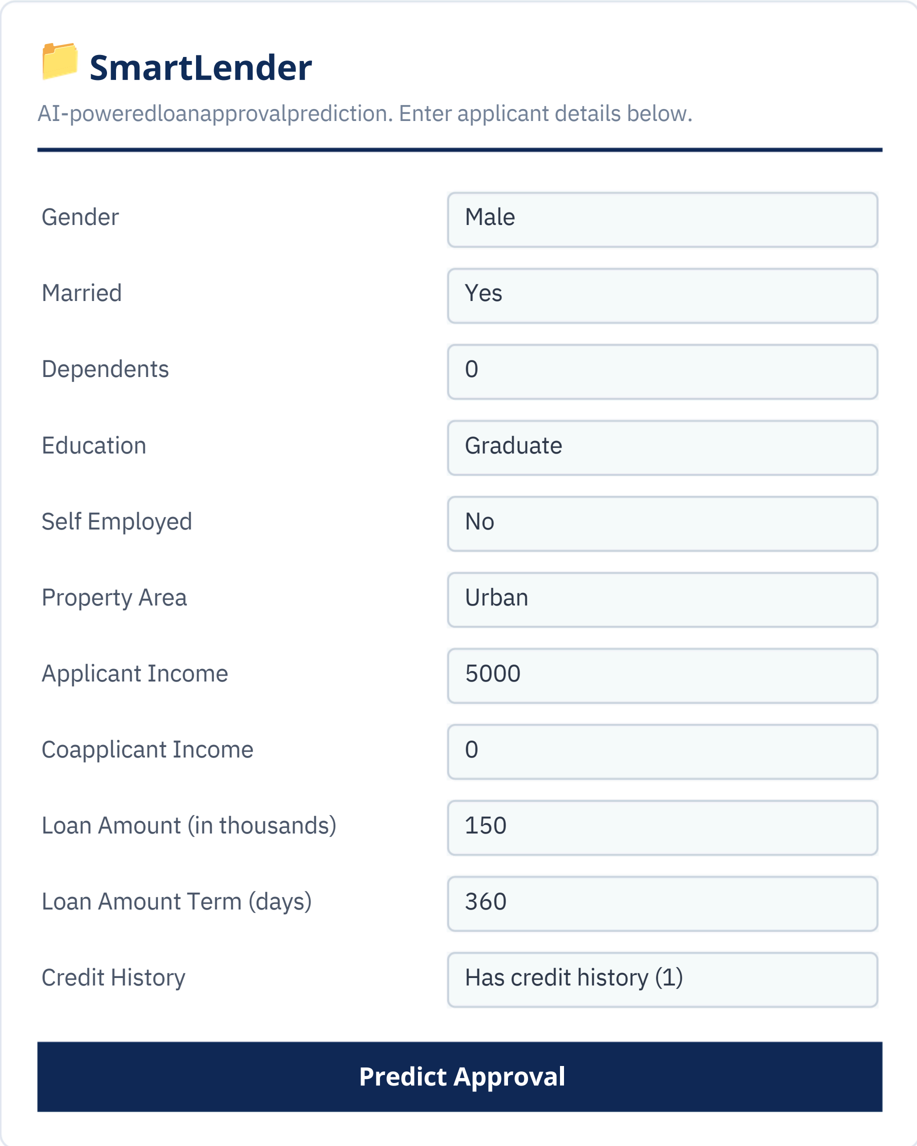

Figure 6: Input form interface

Web Interface – Prediction Output

After submission, the system returns the approve/reject decision along with the tuned XGBoost model's confidence percentage.

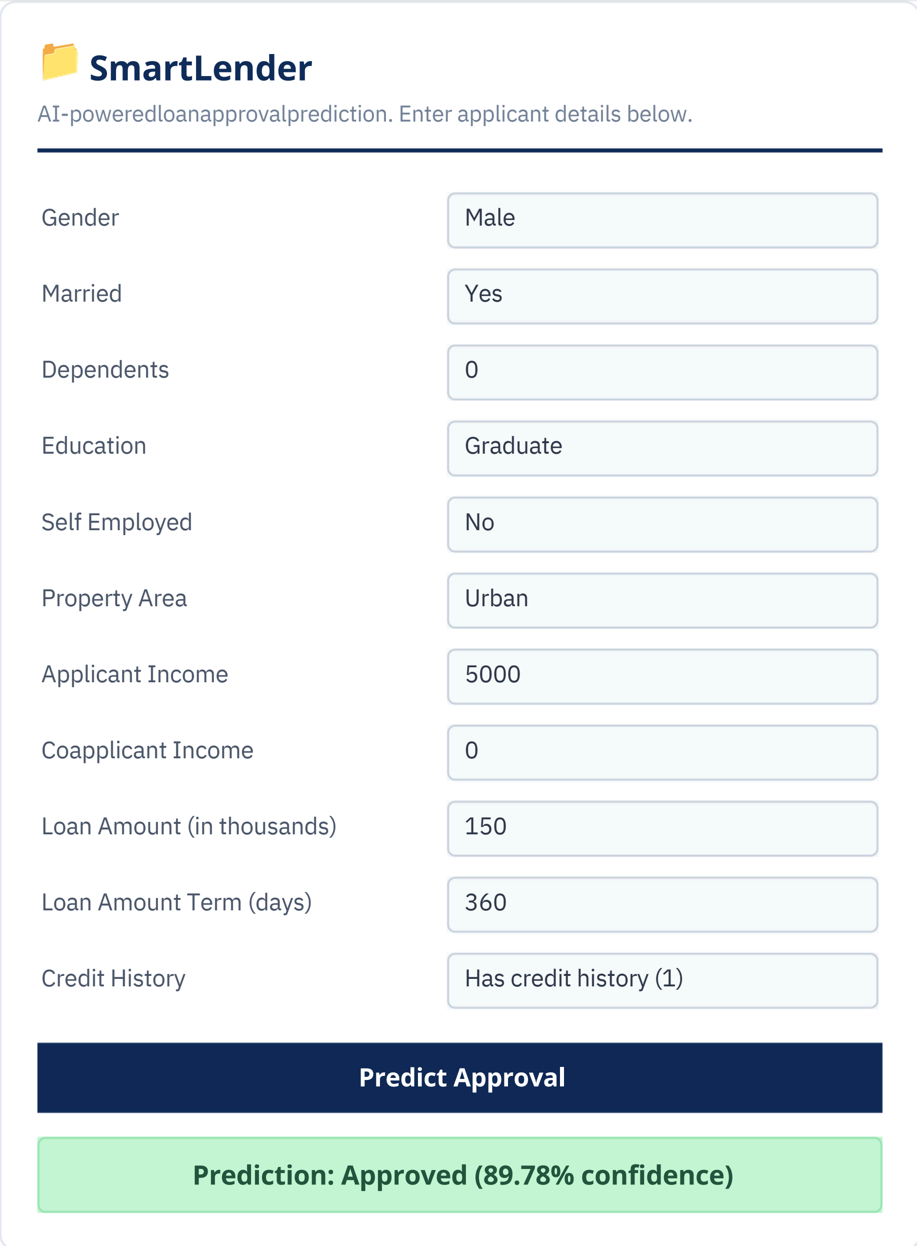

Figure 7: Prediction output display

Conclusion & Future Scope

SmartLender shows that a tuned XGBoost model can automate a meaningful share of the credit approval workload. Future work includes adding explainability for each decision and monitoring the model for drift as new applicant data comes in.